Yerevan State University

Whispering Gallery Mode Resonators

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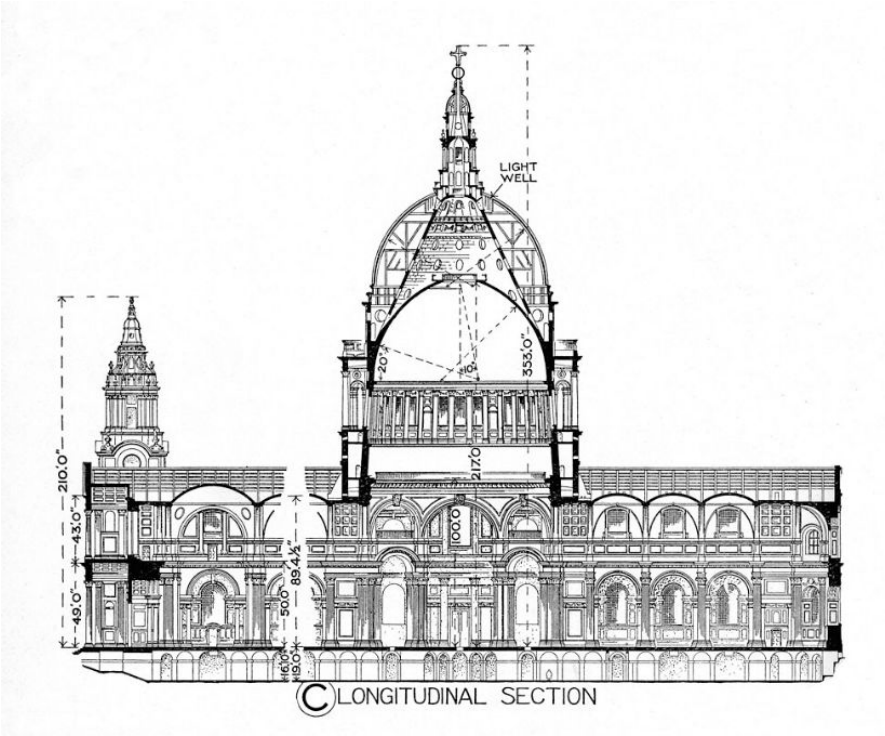
Introduction

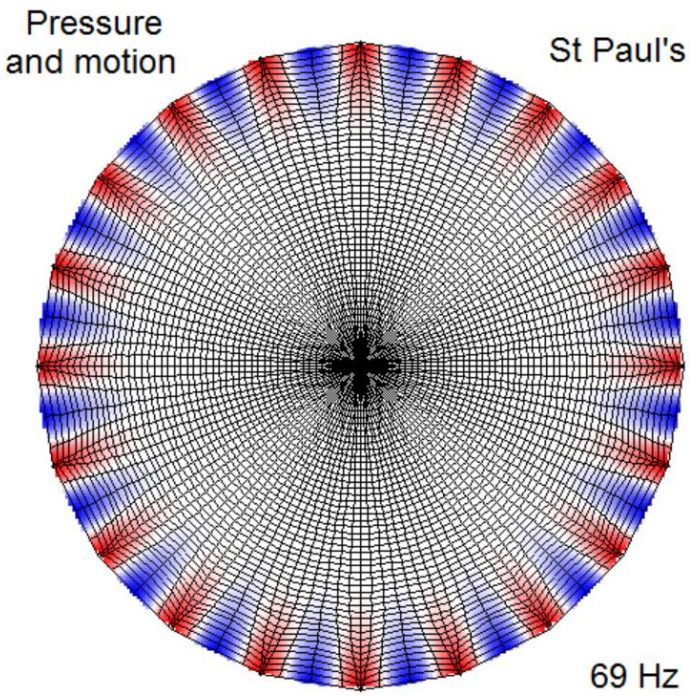
Lord Rayleigh (1878)



St. Paul Cathendral, London

Introduction

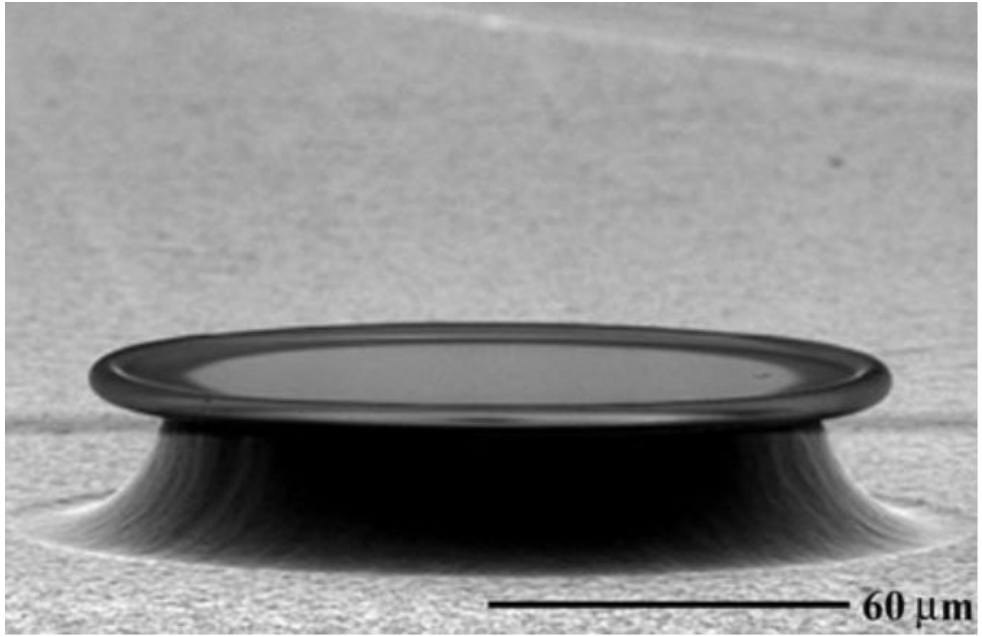
- 69 Hz
- Diameter of 33.7m
- Red and blue represent higher and lower air pressures



WGM Resonators

- Ultra-high Quality Factor (10^8-10^9)
- Small sizes
- Low power requirements
- Super sensitive

Cylindrical,
Spherical,
Spheroidal/toroidal,
Ring etc.



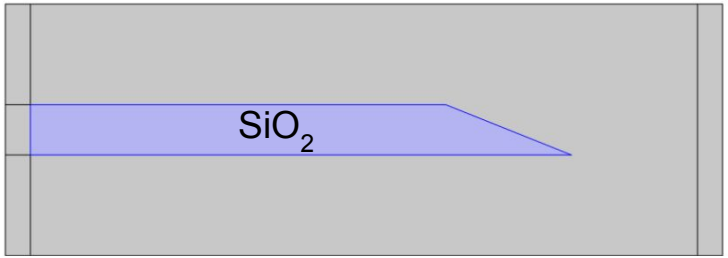
Microtoroid resonator: An SEM image of the microtoroid resonator used in Caltech's biological sensor. (Credit: D K Armani, Caltech).

WGM Resonators

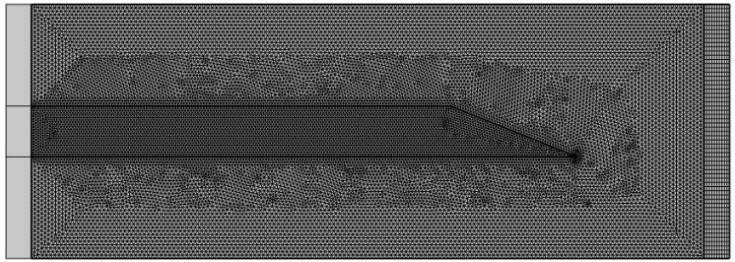
$$N = n - j\kappa$$

$$\alpha = \frac{4\pi\kappa}{\lambda}$$

$$\mathbf{E}(r, \phi, z) = \mathbf{E}(r, z)\exp(-in_m\phi)$$



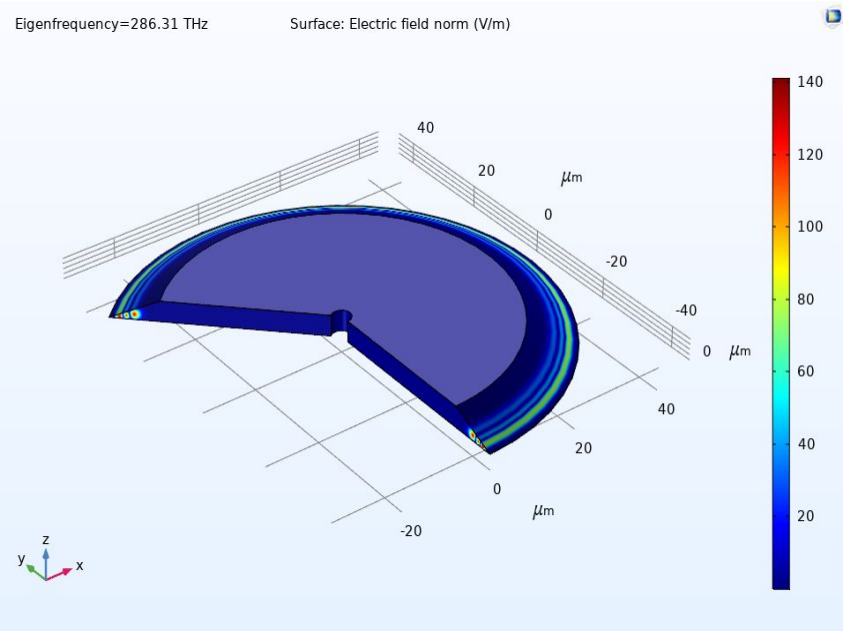
The geometry is surrounded by a Perfectly Matched Layer (PML) to dampen any escaping radiation



WGM Resonator. Idealized System

$Re_{refractive_index} = 1$
$Im_{refractive_index} = 0 \Leftrightarrow \kappa=0$

The electric field distribution of a WGMs with varying mode numbers



WGM Resonator. Idealized System

$$Q_{\mathrm{mat}} = \frac{2\pi n}{\alpha\lambda}$$

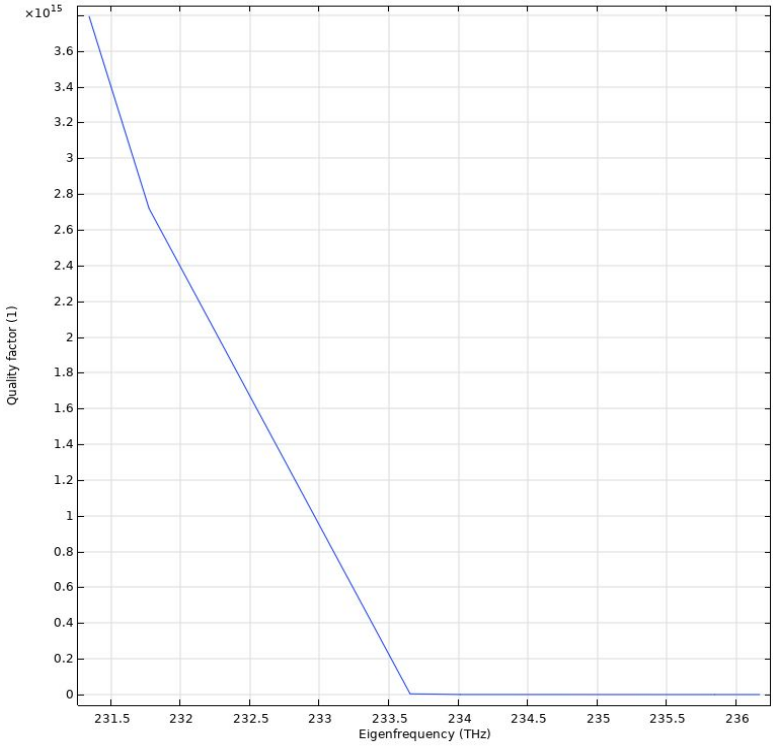
$$\delta_{\mathrm{mat}} = \frac{\omega}{2Q_{\mathrm{mat}}}$$

$$\alpha = \frac{4\pi\kappa}{\lambda} = 0$$

Eigenfrequency λ (THz)

$$\lambda = -j\omega + \delta$$

Ultra High Q Factor (10^{15})

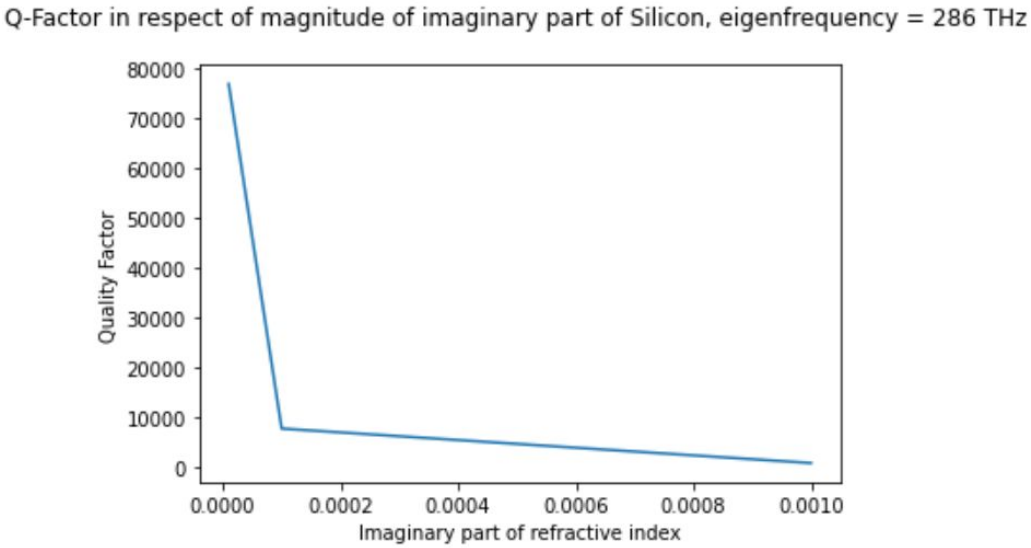


WGM Resonator:Nonzero Values for Imaginary part of Refractive index

Eigenfrequency = 286 THz

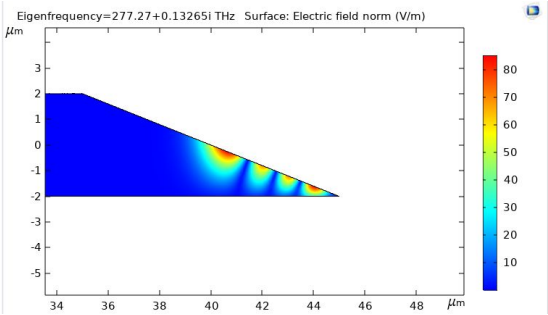
Imaginary part of refractive index varies between 10^{-5} and 10^{-3}.
Data points are 10^{-5} , 10^{-4} , 10^{-3}

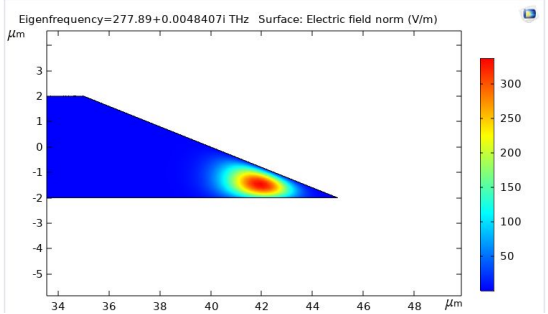
Q-factor drops from 10^{15} to 10^{4}-10^{2}



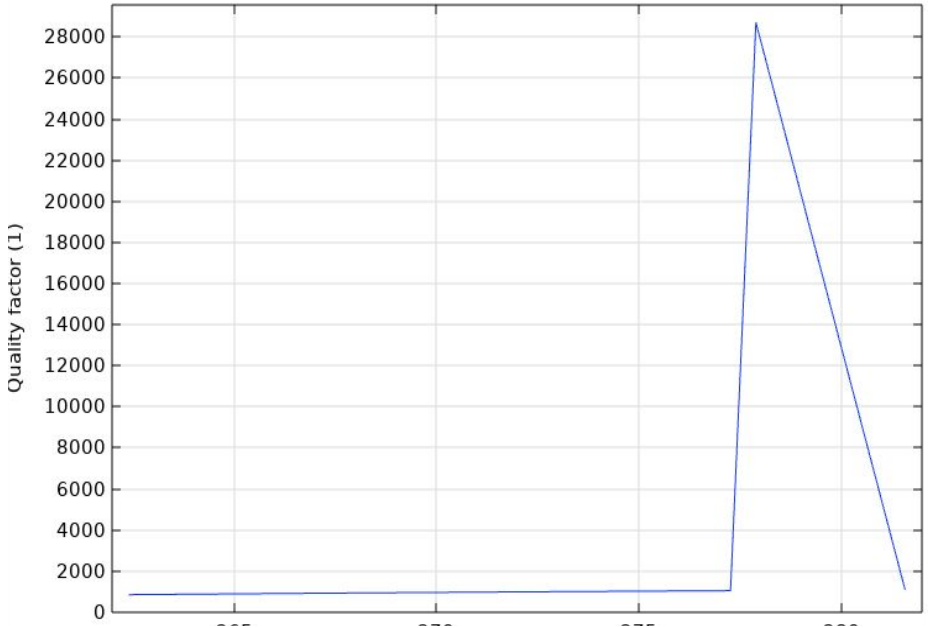
Effect of a Thin Layer of Silver

Layer Thickness = 50 [nm]





Q-factor drops from 10^{15} to 10^{4}



Effect of a Thin Layer of Silver. Conclusion

- Resonator Ohmic heating or other mechanisms
- Change of the Optical properties: alter the refractive index, additional scattering sites
- Capacitance or inductance

Thank You